

Operations Research 2022-2023

Exam Questions

— *Do not turn this page before the official start of the exam!* —

First Name, Surname: _____

Student ID: _____

Examiner: Dr. Barbara Franci

Date/time: Wednesday 25th January 2023 (exam time: 2 hours)

Format: Closed book exam

Allowed aids: Pens, simple (non-programmable) calculator.

Instructions to students:

- The exam consists of 6 questions on 15 pages (excluding the 1 cover page(s)).
- Fill in your name and student ID number **on each page**, including the cover page.
- Answer every question at the reserved space below the questions. If you run out of space, continue on the back side, and if needed, use the extra blank pages.
- Ensure that you properly motivate your answers.
- Write in a readable way. Answers that cannot be read easily cannot be graded and may therefore lower your grade.
- You are not allowed to have a communication device within your reach, nor to wear or use a watch.
- You have to return all pages of the exam. You are not allowed to take any sheets, even blank, home.
- If you think a question is ambiguous, or even erroneous, and you cannot ask during the exam to clarify this, explain this in detail in your answer.
- If you have not registered for the exam, your answers will not be graded, and thus handled as invalid.
- **Good luck!**

The following table will be filled by the examiners:

Question:	1	2	3	4	5	6	Total
Points:	20	15	25	15	10	15	100
Score:							

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Question 1 (20 points)

This question is about the Transportation Problem. The two subquestions are not related.

1. Santa Claus has to assign the elves to the construction of toys before Christmas. In total, he needs 50 millions (ML) of remote cars, 65 ML dolls and 40 ML board games. He has three teams of elves at his disposal to build the toys (independently of being cars, dolls or board games). The first team can build 45 ML toys at the unit price of 20 coins per toy. The second team can build 60 ML toys at the unit price of 25 coins per toy, but they cannot build board games since they do not know the rules. The last team can build 40 ML toys and asks 10 coins for the board games and 12 coins for the cars; however, they are not able to sew so they cannot make the dolls.

Explain how you can model this cost-minimization problem as a Transportation Problem. Define the variables and write the transportation table. You do not need to solve the problem.

2. Consider the Transportation table below. There are 3 supply points (with supplies 12, 7 and 11) and 3 demand points (with demands 8, 9, 13). The table shows the cost of shipping one unit from each supply point to each demand point. We wish to minimize the shipping costs.

		Destination			Supply
		1	2	3	
Source	1	9	10	7	12
	2	3	2	1	7
	3	6	3	2	11
Demand		8	9	13	

Consider the following basic feasible solution: $(S_1, D_1) = 8$, $(S_1, D_2) = 4$, $(S_2, D_2) = 5$, $(S_2, D_3) = 2$, $(S_3, D_3) = 11$. Starting from this basic feasible solution, perform a single pivot of the Transportation Simplex method. After this single pivot, indicate which variables are now basic, their values, and the cost of the current solution. Comment on whether the solution is optimal.





Queuing process.

1. What is the expected number of patients in the system at any point in time?
2. Suppose the clinic has just opened. How long does it take before all 3 dentists are treating a patient?



Question 3 (25 points)

Simplex method for linear programming.

Consider the following linear program:

$$\begin{array}{llllll} \text{maximize} & Z = & 3000x_1 & + & 5000x_2 & \\ \text{subject to} & & x_1 & + & x_2 & \leq 12 \\ & & 7x_1 & & & \leq 70 \\ & & & & 3x_2 & \leq 18 \\ & & 10x_1 & + & 20x_2 & \leq 160 \\ & & x_1 \geq 0 & & x_2 \geq 0. & \end{array}$$

1. Use the graphical method to find the optimal solution. Explain how you find it and give your solution as $x_1 = \dots, x_2 = \dots, Z = \dots$
2. Use the tabular form of the simplex method to solve the linear program. Explain how you choose your *pivots* and give your solution as $x_1 = \dots, x_2 = \dots, Z = \dots$





Question 4 (15 points)

Flipping coins.

Consider repeatedly flipping a fair coin. The results of these flips, H's and T's, are listed in a *sequence*, one by one. A finite string of consecutive flips is called a *code*.

1. We now are interested in the probability q that the code HHH appears sooner than TTT in the sequence of flips. Model the problem by a *transition diagram* (graph) and write a system of equations (built from your diagram) using the variables p_x for the probability of reaching HHH sooner than TTT when starting at state x . Clearly indicate which p_x corresponds to the probability q and compute it.
2. Calculate the expected number of flips until the code HHH occurs for the first time.





Question 5 (10 points)

Game Theory: Bimatrix Games.

Find *all* the equilibria and their values for the matrix game given by:

$$\begin{pmatrix} (0, -1) & (2, 0) \\ (1, 2) & (0, 1) \end{pmatrix}$$

Player 1 plays rows, Player 2 plays columns; they both want to maximize their profit.



Question 6 (15 points)

Markov chains.

Consider the discrete Markov chain with state space $\{1, 2, 3, 4, 5\}$ and transition matrix given by

$$P = \begin{pmatrix} P_{11} & P_{12} & P_{13} & P_{14} & P_{15} \\ P_{21} & P_{22} & P_{23} & P_{24} & P_{25} \\ P_{31} & P_{32} & P_{33} & P_{34} & P_{35} \\ P_{41} & P_{42} & P_{43} & P_{44} & P_{45} \\ P_{51} & P_{52} & P_{53} & P_{54} & P_{55} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0.1 & 0.1 & 0.8 & 0 & 0 \\ 0 & 0.5 & 0.3 & 0.2 & 0 \\ 0 & 0 & 0 & 0.3 & 0.7 \\ 0 & 0 & 0 & 0.8 & 0.2 \end{pmatrix}$$

1. Calculate, for initial state 3, the probability of ever arriving at state 1, and also calculate the probability of ever arriving at state 4.
2. Calculate the unique stationary distribution π that corresponds to initial state 3.
3. For the process starting at state 3, calculate for each transient state s , the expected number of stages that the process will be in that particular state s .







